



Scale of Analysis

RRHS AP Human Geography
2019-2020

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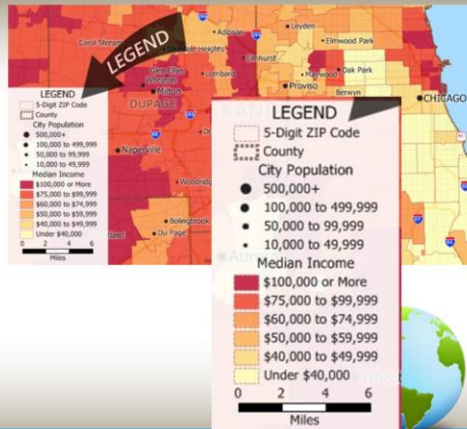
You've Got Data. Now what?

- Determine how to best display data
 - Maps?
 - What kind?
 - At what level of analysis?
 - Charts?
 - Diagrams?



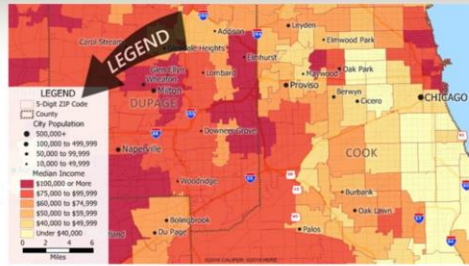
Range of observations

- data being displayed is chunked into ranges.
- determines how it is displayed on the map (shading, altering size (cartograms)).
- Legend provided so user can interpret the patterns.



Scale of Analysis

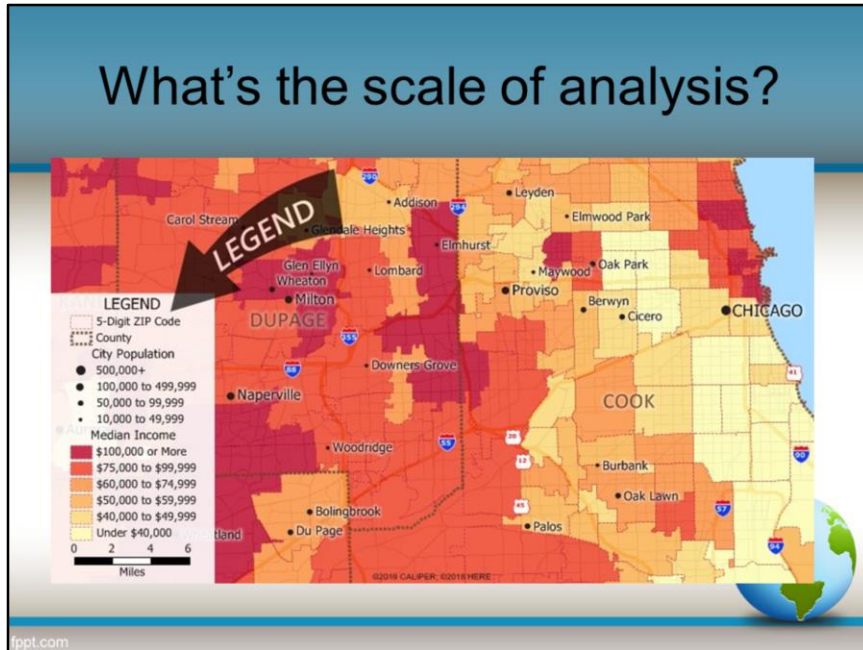
- Not to be confused with “study area”
- Includes both the range of observations in the legend AND
- The geographic unit at which the data is being displayed



The study area is the entire area being studied, while the scale of analysis is the geographic unit at which data is being displayed combined with the ranges of observations found in the legend.

Manipulating the geographic units and the data ranges will impact how the patterns appear on the map (more about that in examples 1 and 2 later).

What's the scale of analysis?



The study area of the map above is the metropolitan area around Chicago, Illinois. (Metropolitan area = large city and surrounding counties)

On the map, you can see that each color represents a range of median income (median means half the people make more, half the people make below. Median = midpoint). The lowest range of median income is under \$40,000, the next darker shade is \$40,000-49,999, the next \$50,000-59,999 and so on and so forth).

The geographic unit at which the data is displayed is also indicated on the map legend – it's the 5-digit ZIP code, or postal code.

What patterns do you see based on this information?

What other information is displayed on the map?

Global Scale

- Unlikely to be used for unit of analysis.
There would be no patterns revealed due to no geographic units unless we were looking at the larger solar system.
- More likely to be the “study area”



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The study area is the larger geographic area about which data is being reported. When we talk about something happening on “a global scale”, what that means is that the occurrence is world-wide or impacts many people in many places.

Regional Scale

- Large chunks of the earth sharing similar physical or human characteristics



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Regions are groups of places/countries that contain multiple countries.

For example, on the map above, North America is Separated from Latin America, yet Mexico and parts of Central America which are located on the North American continent are included with Latin America. The reason is that North America is sometimes referred to as “Anglo-America”, which means English-speaking America, while Latin-America refers to the countries where the residents speak the Latin-based languages of Spanish and Portuguese.

Major world regions are more likely to be used when studying data for the entire world, but in very broad terms. We know that not everyone in North America speaks English, there are many areas, such as the southern U.S. and Quebec in Canada, where Spanish and French are the dominant languages, respectively.

Shared culture, political ties, and history are the main characteristics used to create these large world regions.

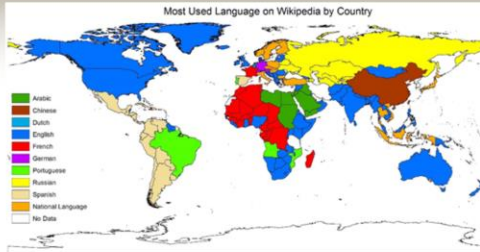
Refer to your Regions maps for examples

May be further sub-divided into smaller regions that are still larger

than all but the largest countries (i.e, the 2nd map of the regions handout).

National Scale

- AKA – country scale.
- Data is displayed by country.



For the national scale to be the appropriate scale of analysis, the study area must be larger than a country. It may be the entire world, or it may be a major world region.

The map above displays the most-used language on Wikipedia by country – this means submissions to Wikipedia are typically in the language shown, rather than the languages being spoken in that place. This map gives us insight into the culture of a country as secondary or common languages typically reflect the history of that place. The study area is “global”, and the scale of analysis is “national level”.

Sub-national Regions

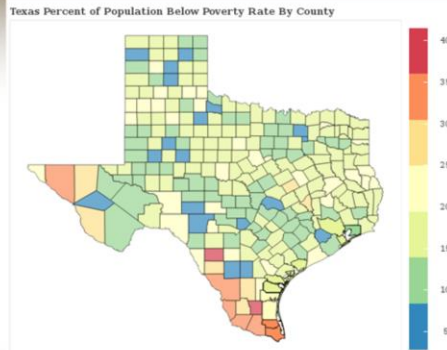
- Smaller regions within a country



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Local Scale

- Smallest geographic units
 - County
 - City
 - Zip code
 - Neighborhood
 - Census tract or block



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When the study area is a state, the most common geographic unit is the county.

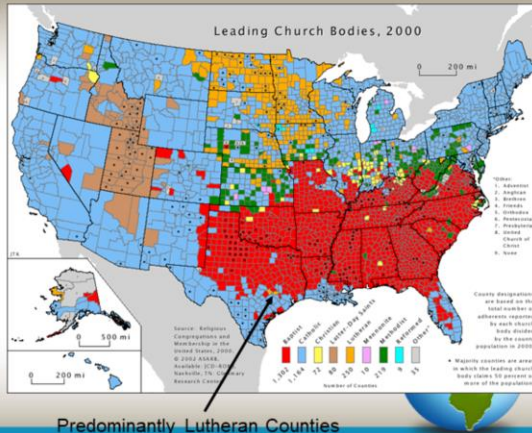
Sometimes will see county level reporting for the entire U.S.

City/metropolitan area information (such as the Chicago metro area map from a few slides ago) data is typically displaced by zip code or census tract.

Patterns and Analysis

The Geographic Method:

- Make observation (what)
- Identify locational patterns (where)
- Analyze underlying characteristics of place (why there)
- Ask additional questions (why care)
- Data collection
- Repeat



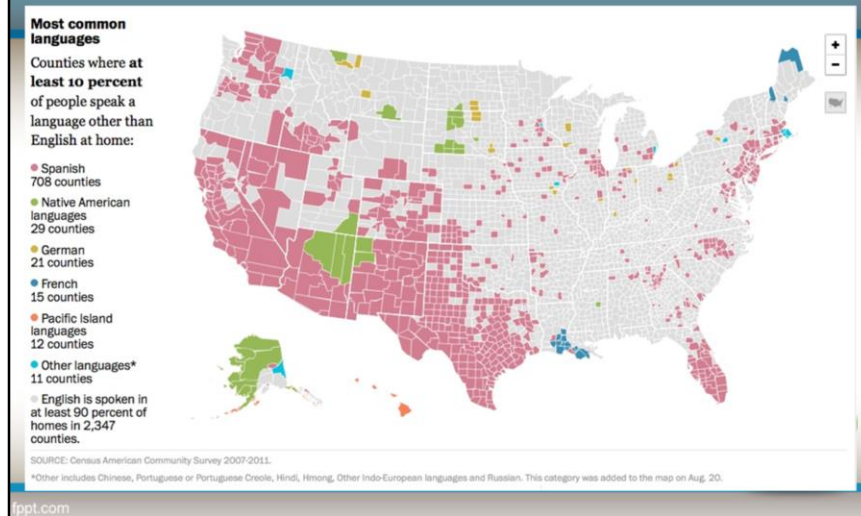
Predominantly Lutheran Counties

In the map on the previous page, where were the highest percentage of people living below the poverty line located? What parts of Texas?

Some patterns are very easy to see, but others are not so clear. Geographers often manipulate the scale of analysis (data reported at the county vs state vs other) in order to find the level at which the pattern is the most clear.

For example, the map above shows the “Leading Church Bodies” in each county. The map makers *could* have used a state level analysis, but many important church bodies would have been lost due to generalization. Instead of the leading bodies per county, it would have only shown the most dominant church participation by branch or denomination by *state*. We would not be able to see the two counties just to the east of us that are predominantly Lutheran (wonder why? – there is a German enclave in Lee and Washington Counties from when several Lutheran groups were pushed out of Catholic Germany). Meanwhile, the Lutheran counties of the Midwest (the orange counties of North Dakota, Minnesota, Iowa, South Dakota, Wisconsin, and Nebraska) mostly were immigrants from the Scandinavian countries of Northern Europe. Had this data been presented state the state level, we likely would not be able to see those relevant details.

Level of generalization

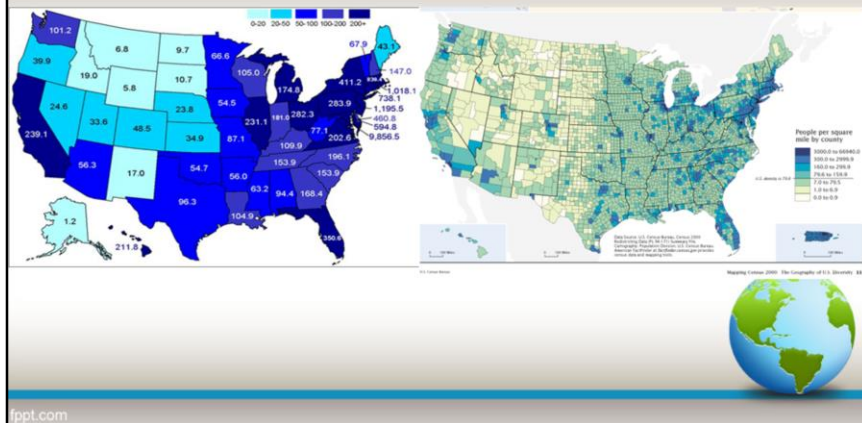


At its core, cartography is about *abstraction*. We don't show data in its raw form; we clarify it in a variety of ways, often by removing things. It simply isn't possible to show every tiny detail! Data and graphics should be *generalized* appropriately to the geographic unit at which the data is being displayed. If we're looking at the predominant languages spoken at home throughout the United States, we are likely to not see any patterns if we display the data at the state level – the predominant language spoken in every state is English. But, if we look at the most common language *other than English* at the county level, distinctive patterns become evident. Notice the Native American languages spoken in Arizona, New Mexico, Utah, Montana, the Dakotas, and Alaska. French is spoken around New Orleans and in Northern Maine due to the historic French colonies in those places. Spanish-speaking counties reflect both the early colonial patterns of the United States as well as proximity to Spanish speaking countries (they are not speaking Spanish in these places due to recent migration, they have been speaking Spanish there since those regions were part of Mexico before the end of the Mexican American War of 1848 when Mexico ceded almost half its land area to the U.S.). Spanish speaking counties in the North East reflect migration from Puerto Rico (a U.S. territory), while Southern Florida is a combination of colonial settlements as well as more recent migration from Puerto Rico, Cuba, and other Spanish speaking countries. As you can see,

recognizing the patterns is only one part of the equation of spatial analysis – examining the history, culture, flows, and other characteristics of the places help to tell us WHY those patterns exist. When you see a pattern, always ask yourself “Why there?” and then continue digging for information.

Example 1

- Population Density (how many people per square mile)

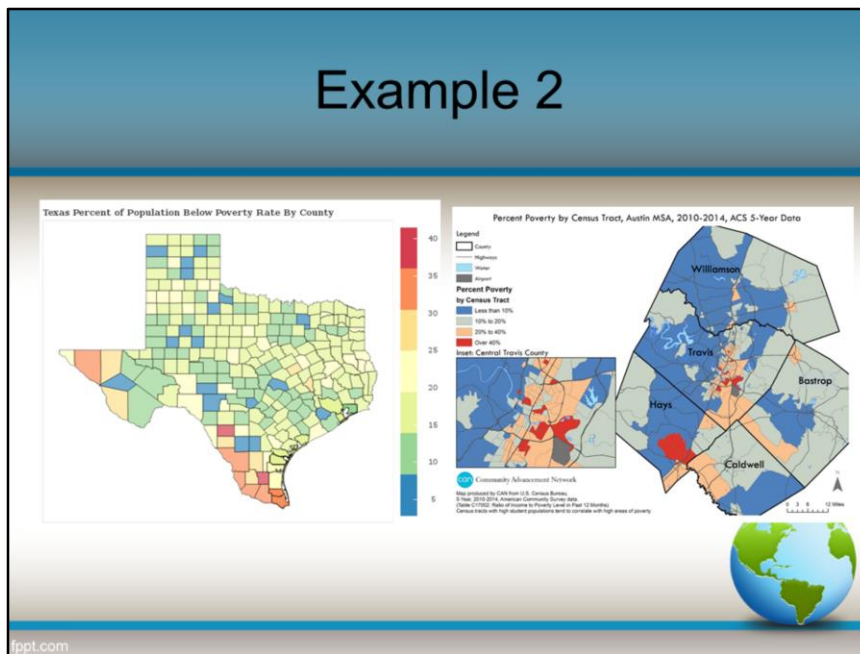


These two maps display the same data, but at different scales of analysis.

The study area for both is the United States. The map on the left displays the data at the state level, while the map on the right displays the data at the county level. While both maps give us a pretty good idea of where population lives within the country, if you look at the 2nd map, you can see that even within some of the more densely populated states there is significantly uneven population distribution – cities are far more densely populated than rural areas, and you can identify the location of the cities easily over the more open countryside.

Question (just think about it) – In what cases would you prefer the state-level data over the county level? The county level over the state?

Example 2



The size of the area also determines the patterns we see. If we're zoomed out too far, too many details are generalized and the patterns do not reveal the whole picture. Zoom in further, and we can see more clearly what the patterns are – but you can zoom in too far, as well.

For example, in the maps above, the data displayed is showing the percent of people living below the poverty level. On the map on the left, we can clearly see that counties in West Texas and the Rio Grande Valley (The Valley) have relatively high proportions of residents living below the poverty line.

The map on the right is zoomed in – but it is showing Travis and surrounding counties (which on the map on the left have relatively low % of the population living below the poverty line) and displays the data by census tract. Zooming in this far, we miss out on the pattern of poverty in the state as a whole, but we do see the pockets of relatively high poverty within the shown counties. The inset on the map to the right shows *just* Travis county and census tracts with a high rate of poverty that get washed out in the more zoomed out maps.

To sum it up – how data is displayed will impact how they are analyzed and the

information learned from them. Every time you see a map, ask yourself – what is the study area, and what is the unit/scale of analysis? How do these two decisions made by the map maker influence how I see the information being displayed?